

Virtual Reconstruction of a the Ambisonic Concert Hall of the Royal Danish Academy of Music

Anders Riddersholm Bargum *

Devansh Kandpal †
Jesper Andersen ‡Oddur Ingi Kristjansson ‡
Stefania Serafin ¶

Simon Rostami Mosen §

Aalborg University Copenhagen
Royal Danish Academy of Music

Figure 1: **Left:** A part of 360° image taken at the Ambisonic concert hall of the Royal Danish Academy of Music. **Right:** A VR rendering of said concert hall.

ABSTRACT

In this paper we present a simulation and validation of the Ambisonic concert hall at the Royal Danish Academy of Music (RDAM), which was inaugurated on September 19th, 2019 in Copenhagen.

Validation with experts showed that the simulated ambisonic room was perceived to match the acoustic- and spatial attributes of its real counterpart.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Virtual simulations of real concert halls are useful from different perspectives. Simulating the acoustics of a space allows to better understand its sonic characteristics and the relationship between the physical design and the way sound propagates.

Moreover, composers who want to write music for such spaces can test their spatialization and sound rendering techniques in the virtual space, before trying it in the real space. This is particularly useful for concert halls that are hard to access, due for example to high demand or more recently to the issues generated by the COVID-19 pandemic.

In this study we are interested in simulating the Ambisonic concert hall of the Royal Danish Academy of Music. The Hall was inaugurated on September 19th 2019. The Ambisonics system in the immersive hall comprises a staggering 48 channels with 44 speakers

(Dynaudio LYD) and 4 subwoofers (Dynaudio 18S). The speakers are installed to form a hemisphere as accurately as possible.

We describe the simulation of the space and report on how the concert hall simulation has been evaluated against its real counterpart to validate the quality of the simulation.

2 PREVIOUS WORK

There are several perceptual parameters associated with room acoustics that need to be considered when obtaining a perceptually realistic simulated auralisation in VR [11]. Capturing and reproducing 3D sound independently of listener and speaker configuration has traditionally been elicited by an Ambisonic system, which has the possibility of providing a full sphere experience with smooth panning [1, 2].

Early scientific research on ambisonics was conducted in the early 1970s by, among others, Gerzon and Felgett [3, 6]. Ambisonics aims at generating sound fields in all directions where sound fields are decomposed into spherical harmonics, enabling a true spatial sound experience. While ambisonics was proposed as an alternative to quadrophonic speaker setups for spatial sound in the early 1970s [3], it has come to the fore recently as a tool for exploring different spatialisation techniques, with the advent of powerful digital signal processing (DSP) technologies. 1st order ambisonics, also called B-format, is encoded using plane waves for a set of 4 speaker channels (W, X, Y, Z). Higher-order ambisonics (HOA) encode audio information in multiple channels, hence offering a high level of spatial sound resolution and precise acoustic sweet spots. This is encoded using the equation of spherical harmonics, given as [1]:

$$S(t; \theta, \delta) = \sum_{l=0}^{\infty} \sum_{m=-l}^l B_{lm}(t) Y_{lm}(\theta, \delta) \quad (1)$$

Where $s(t)$ represents a mono signal to be encoded and (θ) and (δ) represent the azimuth and elevation panning of the mono signal respectively [1]. $B_{lm}(t)$ correspond to the HOA channels, and contain the traditional (W, X, Y, Z) channels, where $Y_{lm}(\theta, \delta)$ represents the spherical harmonics [1]. The wave-equation for each

*abargu17@student.aau.dk

†dkandp20@student.aau.dk

‡okrist17@student.aau.dk

§e-mail:srosta17@student.aau.dk

¶Jesper.Andersen@dkdm.dk

¶e-mail:sts@create.aau.dk

channel at the given order, with its corresponding coefficient/weight m , can as an example be found and implemented using general tables like the one in [1]. As equation 1 suggests, signals encoded for ambisonics do not feed any speakers directly, but rather carry information pertaining to a sound field. The Ambisonic sound can then be decoded into any speaker array bigger than the number of channels associated with the given order, and in the case of this project, thereafter rendered into binaural sound by convolving it with relevant head-related transfer functions (HRTFs), making it fit for use in general audio applications [8].

Since calculating the sound source into its equivalent spherical harmonic is both frequency and distant dependent, Ambisonic systems are tailored for a listening experience in a certain sweet spot [4]. This makes translation within the Ambisonic field hard and may create additional problems for the traditional mode-matching decoder, which is only suitable for regular loudspeaker arrangements covering the full sphere as described in [5]. Luckily, the ambisonic concert hall at RDAM, which this project aims to simulate, is comprised of a 48 speaker system aiming to represent a full sphere, allowing for high order, high-resolution reproduction.

When simulating acoustic spaces, a range of different parameters should be accounted for. Serafin et. al [14] outlined how a typical system for binaural rendering and auralisation consists of different components to render realistic audio. VR simulations of real spaces,

in Paris [9, 12] or to recreate spaces that do not exist anymore [13] or for multimodal perception experiments [7].

By simulating and auralising the Ambisonic room at RDAM, a realistic virtual audio-visual listening environment was created.

3 DESIGN AND IMPLEMENTATION

Considering all the different elements presented in [14], the simulation of the Ambisonics concert hall can be split into the following components: VR scene, source, sound propagation, binaural rendering, listener position tracking and headphone equalisation.

The concert hall was first captured using a 360 camera visually replicated as a 3D environment in Unity to realistically depict the room from RDAM. As can be seen in Figure 1, spatial proportions and visual aesthetics were replicated accurately. Moreover, speaker positions in the 3D environment were placed according to the location specifications of the ambisonic system see Figure 2. The concert hall and its real counterpart are seen below on Figure 1.

For the simulation, the rendered Ambisonic audio can be contemplated as the sound source in the scene. To match the ambisonic rendering from the concert hall at RDAM, a similar encoding and decoding procedure used by the conservatory, was implemented:

1. Mono or stereo audio files were encoded to 5th order through a script made in MATLAB.
2. Speaker positions from the concert hall at RDAM were loaded to the script. As the real Ambisonic system includes 4 optional movable subwoofers, 4 speakers representing these were removed from the speaker layout.
3. The Ambisonics encoded audio was decoded using mode matching decoding (MMD) with respect to the speaker positions. MMD is the pseudoinverse of the encoding matrix used, as it is the traditional decoding matrix used for full-sphere systems [5].
4. Individual .WAV files, corresponding to the audio signal at each individual speaker, were exported to be played back from each individual virtual speaker in the simulation.
5. A SOFA file, by name FHK_HRIR_2354¹, with a minimal coloration of the sound was found and the simulated Ambisonic audio was rendered binaurally using Steam Audio in Unity <https://valvesoftware.github.io/steam-audio/>.

To resemble realistic sound propagation, impulse responses (IR) from different locations of the concert hall were recorded and used for convolution with the sound source to capture and realistically recreate the reverberation present in the concert hall. The room was excited using a sine sweep while the response of the room was recorded. The IR recording was deconvolved to estimate the real response of the room. Finally, each speaker was, depending on its position, routed to one of four IRs corresponding to an area of the room. Convolution reverb was used as room reverberation method, merely for its realism.

The rendered Ambisonic audio was summed into a binaural output using HRTF, from the previously mentioned SOFA file, at the position of the listener in the simulation. The position of the listener was tracked using the VR head-mounted display (HMD) Oculus Rift, which allows for 6 degrees of freedom.

4 EVALUATION

The similarity between the auralisation and the room was evaluated using a two-fold validation test. Firstly, 4 experts affiliated with the tonmeister education at RDAM, 2 students and 2 teachers, tried the

¹http://www.sofacoustics.org/data/sofa_api_mo_test/

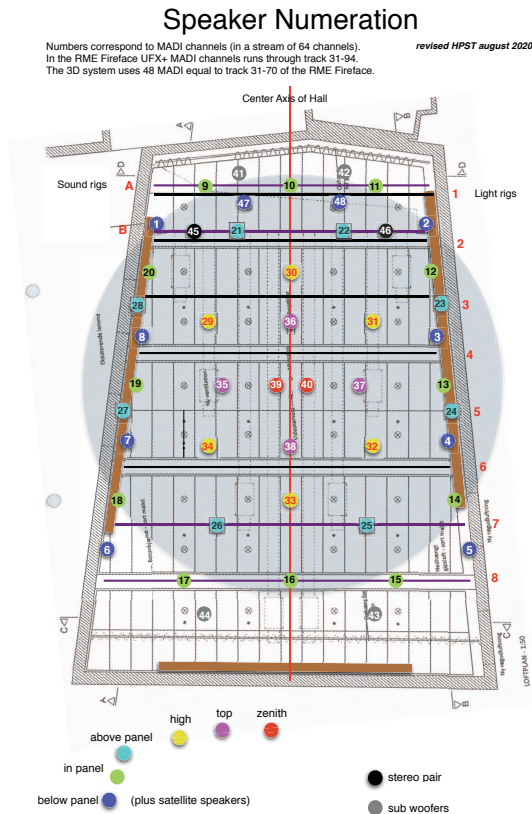


Figure 2: A schematic representation of the concert hall where the position of the speakers is outlined.

both visually and acoustics, have been previously seen. As and example, they were created to simulate the Notre Dame Cathedral

VR simulation, while being in the actual concert hall. Using informal go-along interviews, opinions on aspects such as reverb, tonal coloration, localisation and head-tracking, were gathered. Thereafter, a subjective listening test, evaluating 8 parameters related to the acoustic properties of the room, was carried out. Inspired by the procedure of [10], 13 participants with knowledge and experience in the field of sound production or acoustics were asked to rate the similarity of audio samples from the real room and the simulation, according to eight perceptual attributes. For a more elaborate description of the given perceptual attributes see p. 8 in [10]:

- **Reverberance:** How the sound decay is perceived, where longer decay relates to more reverberance.
- **Clarity:** Refers to which degree distinct element stand out from each other, where high clarity means individual notes stand out from each other.
- **Distance:** Perceived distance from the listener to the sound source.
- **Tonal balance:** Refers to how 'flat' the frequency response of the sound is.
- **Colouration:** Whether the sound is modified compared to the original sound source.
- **Plausibility:** How reasonable the sound is compared to the location.
- **Source Width:** The perceived width of a sound image. How narrow or wide the sound source is.
- **Listener Envelopment:** How balanced the spatial distribution is of the reverberant sound. Higher envelopment refers to more uniform distribution while low envelopment is more narrow/localised.

While 'blindfolded', the participants responded which of the sound files were most affected by the given parameters, using a slider from -50 to 50, where negative values related to the sound file on the left, positive values to the sound file on the right, and 0 indicating no perceived difference between the room and the simulation. The evaluation tool created can be seen in Figure 3.

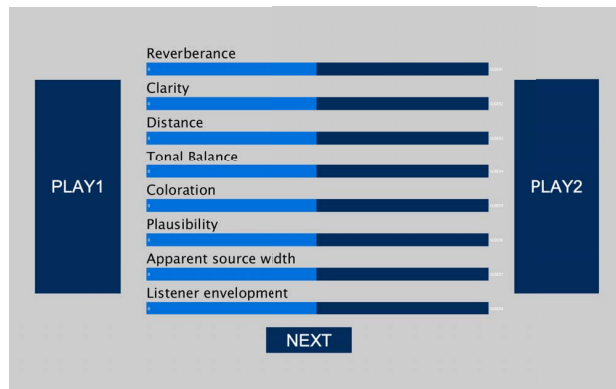


Figure 3: Room validation test was created in Processing.

The two sounds were randomly switched between the left and the right side to minimise response bias. The participants could press 'Play1' and 'Play2' as often as needed before rating each parameter. There were four scenes in total, each with audio samples of different character (Drum n Bass Loop, House Drum Loop, Male Speech, and Choir). To ensure an even comparison, the sound files were without any initial reverberation or room characteristics and each sound source was located at 0° elevation and azimuth, both in the recorded sound as well as the simulated sound.

5 RESULTS

Five attributes (plausibility, colouration, tonal balance, distance and clarity) were considered to be perceived equal, with an individual median attribute difference of 3, 6, 2, 5 and 2 units respectively. The audio samples from the real room were overall considered to be more reverberant than the same samples played through the auralisation, while the sounds of the real room simultaneously were experienced to be wider and have a more uniform distribution of reverb (18 and 30 units difference in favour of the real room). Since several aspects can affect the perceived 'colouration' and 'tonal balance' of the room, such as microphone choice for measuring the room, headphone equalisation, as well as SOFA file and HRTF data for binaural rendering, the small difference in these attributes must be seen as successful.

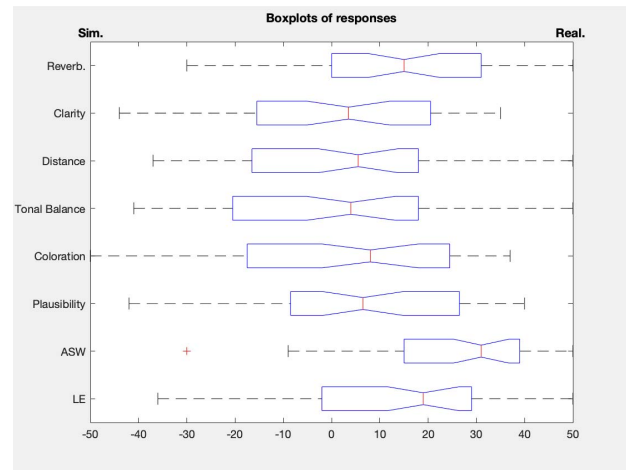


Figure 4: Subject results on similarity of measured and simulated ambisonic room auralisations.

However, looking at the results in Figure 4, it is additionally experienced that the box-limits representing the 25% and 75% quartiles, and the dotted lines, illustrating the highest and lowest data points, are widely distributed, placed at "extreme" values, with an additional outlier in the 'ASW' attribute. This can mean that even though the results hint towards an auralisation that matches the timbral and acoustical colouration of the real room, participants did not seem to agree on the different attributes, creating less valid and reliable conclusions. Which exact attributes to use for interpretation have furthermore been revised after the listening test, as the big difference rating of 'ASW', compared to other attributes, emphasises a flawed testing procedure. The samples from the real room were recorded using two omnidirectional microphones spaced one meter apart, making its recording unnaturally wide compared to a binaural rendering. Thus, the results for the 'ASW' parameter have been excluded from the final interpretation. Additionally, the lack of reverberation of the auralisation was being accounted for by tweaking the convolution reverb used.

Overall, the room validation results match the subjective opinions of the experts, who predominantly had positive feedback. Based on their feedback, the simulation was well received and believed to be very accurate in terms of head-tracking and acoustic reproduction (reverb, material dampening, source width). When being exposed to sound both through the simulation and the actual Ambisonic system after each other, it was furthermore noted that the simulation was 'spot on' regarding localisation. It was, however, uttered that the limited FOV in the HMD, narrowed and affected the perceptual resolution of the sound, making the experts perceive the FOV within the virtual environment to replicate a wider angle than it in fact did. In other words, if the FOV of the headset was 60° (-30°, 30°) with the user looking straight forward (0°, 0°), sounds at -30° and 30° were experienced to be spaced at -90° and 90°. While this was a perceptual effect and not related to the audio engine, it emphasises the importance visuals have when perceiving and recognising sounds and that a change in visuals might affect the perceived audio [15].

6 DISCUSSION

As the results show, the simulated room was experienced to match its real counterpart on several acoustic parameters. However, as the process of representing Ambisonics binaurally involved multiple steps, an error at some point in the signal chain could have caused a non-ideal auralisation, weakening the evaluation. Furthermore, Ambisonics as a precision tool and the given decoding method used should be discussed. While the 5th order encoding used in this study, allowed for high spatial resolution, the smoothness of an Ambisonic system, in general, is accomplished by "giving up on the accurate localisation achievable with a single active loudspeaker", as stated by Frank et. al [5]. As mentioned before, Ambisonics simply represents sound in its given direction as a 'soundfield' rather than a complete position, making the precision of placement, dependable on the density of the speaker layout. Combining this with the use of MMD decoding and the fact that the Ambisonics room of RDM does not represent a complete sphere, localisation difficulties as well as loudness- and source width fluctuations of the auralisation, could affect the use of such simulation.

Overall the evaluation lacked a higher number of participants, which implies a high variance of the results.

7 CONCLUSION

On basis of this study, it is concluded that a realistic simulation of the RDM concert hall was successfully created in VR. It is, furthermore, found that binaurally rendering Ambisonic audio in the simulation, faithfully recreates the Ambisonic sound system as well as acoustic attributes of the real concert hall.

Several directions are possible, for example simulating the reverberation of the room using ray tracing algorithms or custom made reverbs such as feedback delay networks or instead of convolution reverb.

In the future, these simulations will be used by composers at RDM interested in writing music for this space. By having access to a simulation, they are able to test their sound spatialization algorithms remotely before physical access to the space is allowed.

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